LAPORAN TUGAS

PEMROGRAMAN GUI 1 : OPERATOR ASSIGNMENT

Disusun Oleh:

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Operator Assignment

A. Pseudocode

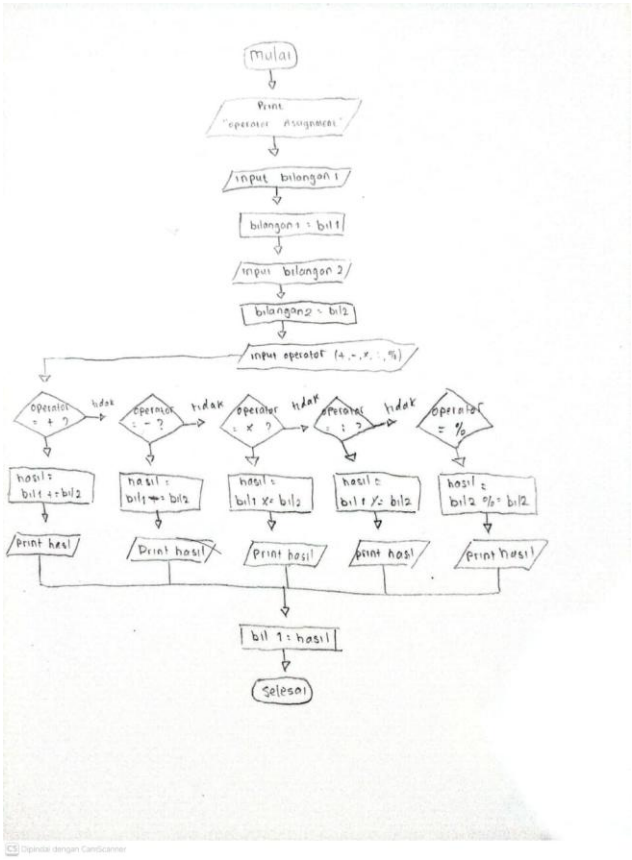
```
Judul :
Operator Assignment
(Program ini berisi sebuah operator logika penjumlahan ,
pengurangan, perkalian, pembagian, dan hasilbagi.)
```

```
Deklarasi :
Var bil1, bil2, hasil ← Integer
Var operator ← string
```

```
Pseudocode :
   1. Mulai
   2. Tampilkan judul "Program Operator Aritmatika"
   3. Minta input bilangan 1
   4. Simpan nilai ke variabel bil1
   5. Minta input bilangan 2
   6. Simpan nilai ke variabel bil2
   7. Minta input operator (+, -, *, /, %)
   8. Simpan nilai operator ke variabel operator
   9. Jika operator adalah "/" atau "%" DAN bil2 = 0 maka
   10.     Tampilkan pesan "Bilangan 2 tidak boleh 0"
   11.     Kembali ke langkah input bil2
   12. Akhir jika
   13. Jika operator = "+" maka
   14.     hasil ← bil1 += bil2
   15. Jika operator = "-" maka
   16.     hasil ← bil1 -= bil2
```

```
17. Jika operator = "*" maka
18.     hasil ← bil1 *= bil2
19. Jika operator = "/" maka
20.     hasil ← bil1 /= bil2
21. Jika operator = "%" maka
22.     hasil ← bil1 %= bil2
23. Tampilkan hasil operasi
24. Set bil1 = hasil   (hasil menjadi nilai bilangan 1)
25. Kosongkan bil2 agar siap input berikutnya
26. Selesai
```

B. Flowchart

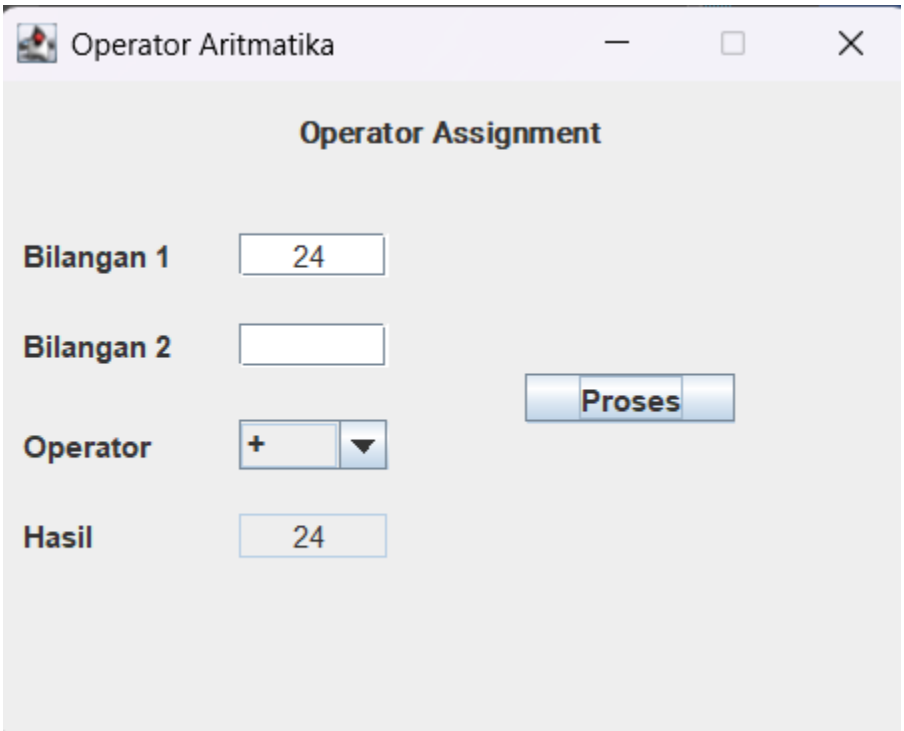


C. Kode Program

```
D. package pekan8_2511532009;
E.
F. import java.awt.EventQueue;
G. import javax.swing.JFrame;
H. import javax.swing.JPanel;
I. import javax.swing.border.EmptyBorder;
J. import javax.swing.JLabel;
K. import javax.swing.JOptionPane;
L. import java.awt.Font;
M. import javax.swing.JTextField;
N. import javax.swing.JComboBox;
O. import javax.swing.JButton;
P. import javax.swing.DefaultComboBoxModel;
Q. import javax.swing.SwingConstants;
R. import java.awt.event.ActionListener;
S. import java.awt.event.ActionEvent;
T.
U. public class Tugas8_2511532009 extends JFrame {
V.
W.     private static final long serialVersionUID = 1L;
X.     private JPanel contentPane;
Y.     private JTextField txtBil1;
Z.     private JTextField txtBil2;
AA.    private JTextField txtHasil;
BB.
CC.    private void pesanPeringatan(String pesan) {
DD.        JOptionPane.showMessageDialog(this, pesan, "Peringatan",
   JOptionPane.WARNING_MESSAGE);
EE.    }
FF.
GG.    private void pesanEror(String pesan) {
HH.        JOptionPane.showMessageDialog(this, pesan, "Eror",
   JOptionPane.ERROR_MESSAGE);
II.    }
JJ.
KK.    /**
LL.     * Launch the application.
MM.     */
NN.    public static void main(String[] args) {
OO.        EventQueue.invokeLater(new Runnable() {
PP.            public void run() {
QQ.                try {
RR.                    Tugas8_2511532009 frame = new Tugas8_2511532009();
SS.                    frame.setVisible(true);
TT.                } catch (Exception e) {
UU.                    e.printStackTrace();
VV.                }
WW.            }
XX.        });
YY.    }
ZZ.
AAA.          /**
BBB.           * Create the frame.
CCC.           */
```

```
DDD.        public Tugas8_2511532009() {
EEE.            setResizable(false);
FFF.            setTitle("Operator Aritmatika");
GGG.            setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
HHH.            setBounds(100, 100, 379, 300);
III.            contentPane = new JPanel();
JJJ.            contentPane.setBorder(new EmptyBorder(5, 5, 5, 5));
KKK.            setContentPane(contentPane);
LLL.            contentPane.setLayout(null);
MMM.
NNN.            JLabel lblJudul = new JLabel("Operator Assignment");
OOO.            lblJudul.setFont(new Font("Roboto", Font.BOLD, 12));
PPP.            lblJudul.setBounds(120, 10, 129, 20);
QQQ.            contentPane.add(lblJudul);
RRR.
SSS.            JLabel lblBil1 = new JLabel("Bilangan 1");
TTT.            lblBil1.setBounds(10, 60, 87, 20);
UUU.            contentPane.add(lblBil1);
VVV.
WWW.            JLabel lblBil2 = new JLabel("Bilangan 2");
XXX.            lblBil2.setBounds(10, 96, 87, 20);
YYY.            contentPane.add(lblBil2);
ZZZ.
AAAA.           JLabel lblOperator = new JLabel("Operator");
BBBB.           lblOperator.setBounds(10, 136, 87, 20);
CCCC.           contentPane.add(lblOperator);
DDDD.
EEEE.           JLabel lblHasil = new JLabel("Hasil");
FFFF.           lblHasil.setBounds(10, 172, 87, 20);
GGGG.           contentPane.add(lblHasil);
HHHH.
IIII.           txtBil1 = new JTextField();
JJJJ.           txtBil1.setHorizontalAlignment(SwingConstants.CENTER);
KKKK.           txtBil1.setBounds(96, 61, 60, 18);
LLLL.           contentPane.add(txtBil1);
MMMM.           txtBil1.setColumns(10);
NNNN.
OOOO.           txtBil2 = new JTextField();
PPPP.           txtBil2.setHorizontalAlignment(SwingConstants.CENTER);
QQQQ.           txtBil2.setColumns(10);
RRRR.           txtBil2.setBounds(96, 97, 60, 18);
SSSS.           contentPane.add(txtBil2);
TTTT.
UUUU.           JComboBox cbOperator = new JComboBox();
VVVV.           cbOperator.setModel(new DefaultComboBoxModel(new String[] { "+",
    "-", "*", "/", "%" }));
WWWW.           cbOperator.setBounds(96, 136, 60, 20);
XXXX.           contentPane.add(cbOperator);
YYYY.
ZZZZ.           txtHasil = new JTextField();
AAAAA.          txtHasil.setHorizontalAlignment(SwingConstants.CENTER);
BBBBB.          txtHasil.setEditable(false);
CCCCC.          txtHasil.setColumns(10);
DDDDD.          txtHasil.setBounds(96, 173, 60, 18);
EEEEE.          contentPane.add(txtHasil);
```

```
FFFFF.
GGGGG.            JButton btnProses = new JButton("Proses");
HHHHH.            btnProses.addActionListener(new ActionListener() {
IIIII.                public void actionPerformed(ActionEvent e) {
JJJJJ.
KKKKK.                    // Validasi input kosong
LLLLL.                    if (txtBil1.getText().trim().isEmpty()) {
MMMMM.                        pesanPeringatan("Silahkan isi bilangan 1 terlebih
   dahulu");
NNNNN.                        return;
OOOOO.                    }
PPPPP.                    if (txtBil2.getText().trim().isEmpty()) {
QQQQQ.                        pesanPeringatan("Silahkan isi bilangan 2 terlebih
   dahulu");
RRRRR.                        return;
SSSSS.                    }
TTTTT.
UUUUU.                    int c = cbOperator.getSelectedIndex(); // operator yang
   dipilih
VVVVV.
WWWWW.                    // Validasi pembagian dan modulus tidak boleh b = 0
XXXXX.                    if ((c == 3 || c == 4) &&
   txtBil2.getText().trim().equals("0")) {
YYYYY.                        pesanPeringatan("Bilangan 2 tidak boleh 0 untuk
   operasi / atau %");
ZZZZZ.                        return;
AAAAAA.                   }
BBBBBB.
CCCCCC.                   try {
DDDDDD.                       int a = Integer.parseInt(txtBil1.getText());
EEEEEE.                       int b = Integer.parseInt(txtBil2.getText());
FFFFFF.                       int hasil = 0;
GGGGGG.
HHHHHH.                       // OPERASI ASSIGNMENT
IIIIII.                       if (c == 0) {
JJJJJJ.                           hasil = a += b;
KKKKKK.                       } else if (c == 1) {
LLLLLL.                           hasil = a -= b;
MMMMMM.                       } else if (c == 2) {
NNNNNN.                           hasil = a *= b;
OOOOOO.                       } else if (c == 3) {
PPPPPP.                           hasil = a /= b;
QQQQQQ.                       } else if (c == 4) {
RRRRRR.                           hasil = a %= b;
SSSSSS.                       }
TTTTTT.
UUUUUU.                       // Tampilkan ke hasil
VVVVVV.                       txtHasil.setText(String.valueOf(hasil));
WWWWWW.
XXXXXX.                       // HASIL MENJADI BILANGAN 1
YYYYYY.                       txtBil1.setText(String.valueOf(hasil));
ZZZZZZ.
AAAAAAA.                      // Kosongkan bilangan 2 agar siap input baru
BBBBBBB.                      txtBil2.setText("");
CCCCCCC.
```

```
DDDDDDD.                    } catch (NumberFormatException ex) {
EEEEEEE.                        pesanEror("Bilangan 1 dan Bilangan 2 harus angka!");
FFFFFFF.                    }
GGGGGGG.                }
HHHHHHH.            });
IIIIIII.
JJJJJJJ.            btnProses.setBounds(211, 117, 84, 20);
KKKKKKK.            contentPane.add(btnProses);
LLLLLLL.        }
MMMMMMM.    }
```

D. Output